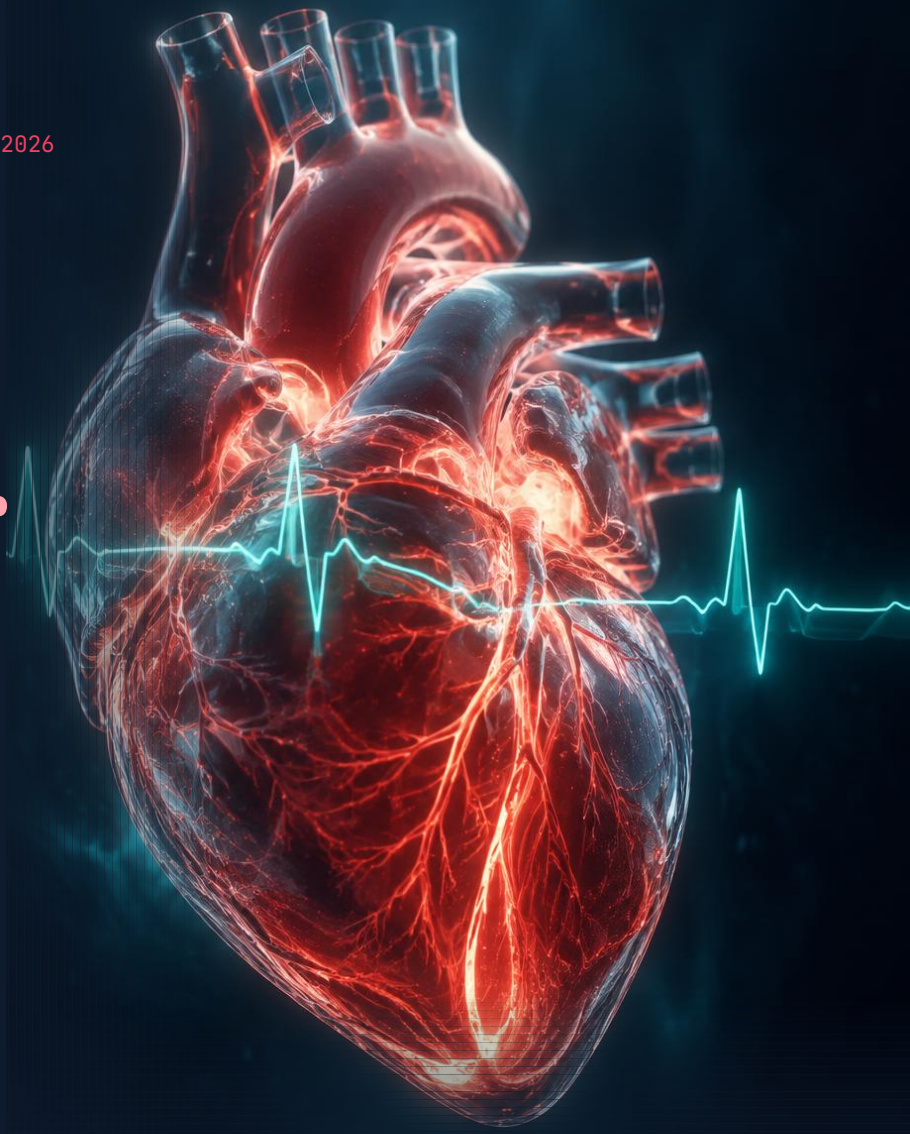
PROJECT REPORT · MACHINE LEARNING · 2026

Cardio *Monitor*

Predicting heart disease from 11 routine clinical signals.



918	11	5	86.96%
RECORDS	FEATURES	MODELS	ACCURACY

Priyam Singh

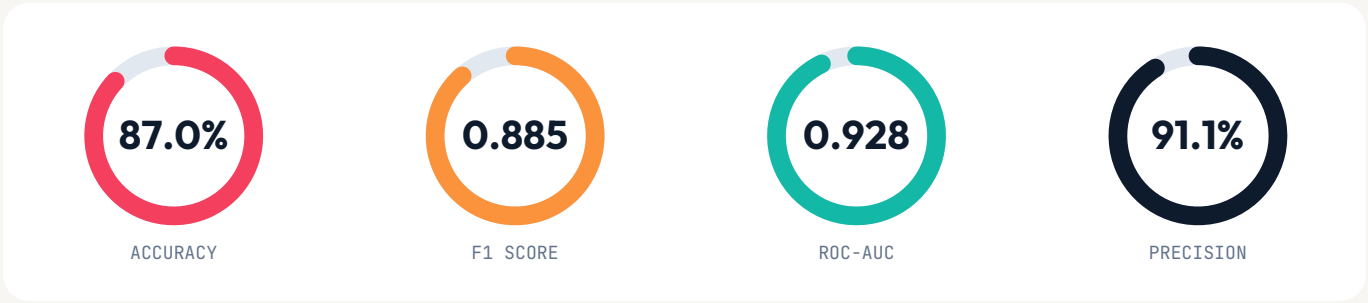
github.com/priyam-10/heart-disease-prediction-ml

LIVE APP

cardio-monitor-ml.streamlit.app

Executive Summary

Cardiovascular disease is the **leading cause of death worldwide**. Catching risk early changes outcomes. This project builds a complete machine-learning workflow — from raw clinical data to a deployed web app — that estimates whether a patient is likely to have heart disease using 11 routine clinical measurements.



Key findings

- **Logistic Regression** won among five models: 86.96% accuracy, 0.885 F1 on 184 unseen patients.
- **ST slope** is the strongest single signal — 82.8% of patients with a Flat slope have heart disease vs 19.7% with Up.
- **Asymptomatic chest pain** is a red flag: 79.0% disease rate, far above atypical angina (13.9%).
- **Exercise-induced angina** raises the rate from 35.1% to 85.2%.
- Risk climbs steadily with age: 34.4% (28–40) → 72.9% (61–77).
- **172 cholesterol** readings (18.7%) were impossible zeros and were repaired before training.

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The Dataset

The **Heart Failure Prediction** dataset (`heart.csv`) combines clinical records of **918 patients**, each described by 11 attributes and a binary target. There are **no missing values and no duplicate rows**.

918	12	508	410
ROWS	COLUMNS	HEART DISEASE	NORMAL

CLASS BALANCE

55.3% disease | 44.7% normal

Feature dictionary

FEATURE	TYPE	DESCRIPTION
Age	Numeric	Age of the patient in years
Sex	Categorical	M / F
ChestPainType	Categorical	TA · ATA · NAP · ASY (asymptomatic)
RestingBP	Numeric	Resting blood pressure (mm Hg)
Cholesterol	Numeric	Serum cholesterol (mg/dl)
FastingBS	Binary	Fasting blood sugar > 120 mg/dl
RestingECG	Categorical	Normal · ST abnormality · LVH
MaxHR	Numeric	Maximum heart rate achieved
ExerciseAngina	Categorical	Angina induced by exercise (Y / N)
Oldpeak	Numeric	ST depression induced by exercise
ST_Slope	Categorical	Slope of peak exercise ST segment
HeartDisease	Target	1 = heart disease, 0 = normal

Summary statistics (raw)

STAT	AGE	RESTINGBP	CHOLESTEROL	MAXHR	OLDPEAK
mean	53.5	132.4	198.8	136.8	0.9
std	9.4	18.5	109.4	25.5	1.1
min	28.0	0.0	0.0	60.0	-2.6
50%	54.0	130.0	223.0	138.0	0.6
max	77.0	200.0	603.0	202.0	6.2

Red minimums of 0 for blood pressure and cholesterol are physiologically impossible — see Data Cleaning.

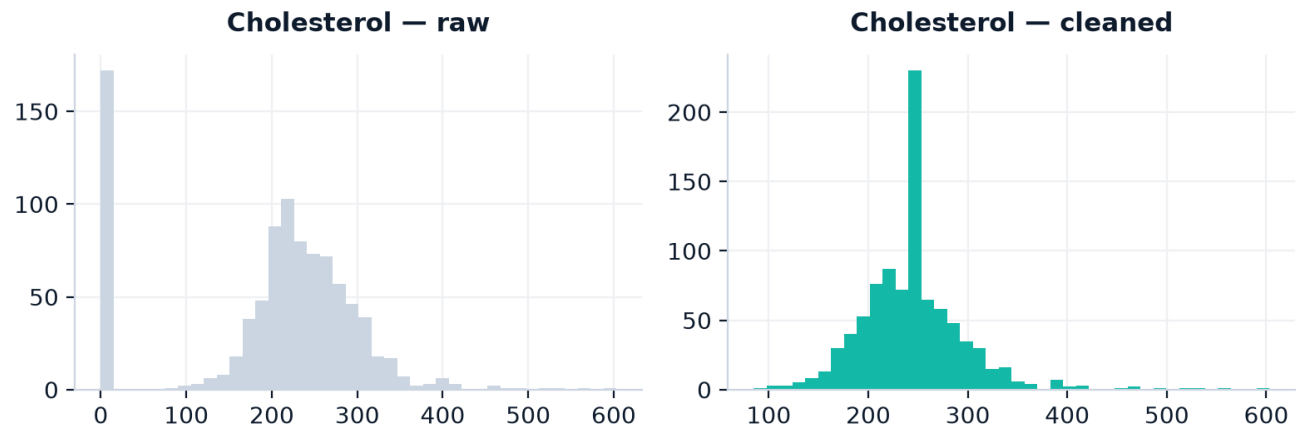
Data Cleaning

A quick look at `describe()` exposed a hidden problem: zeros used as placeholders for missing measurements. A living patient cannot have 0 mg/dl cholesterol or 0 mm Hg resting blood pressure.

172	1	0
ZERO CHOLESTEROL VALUES	ZERO RESTING BP VALUE	NULLS · DUPLICATES

Fix: mean imputation from valid readings

```
ch_mean = df.loc[df['Cholesterol'] != 0, 'Cholesterol'].mean()   # 244.64
df['Cholesterol'] = df['Cholesterol'].replace(0, ch_mean)
resting_bp_mean = df.loc[df['RestingBP'] != 0, 'RestingBP'].mean()  # 132.54
```

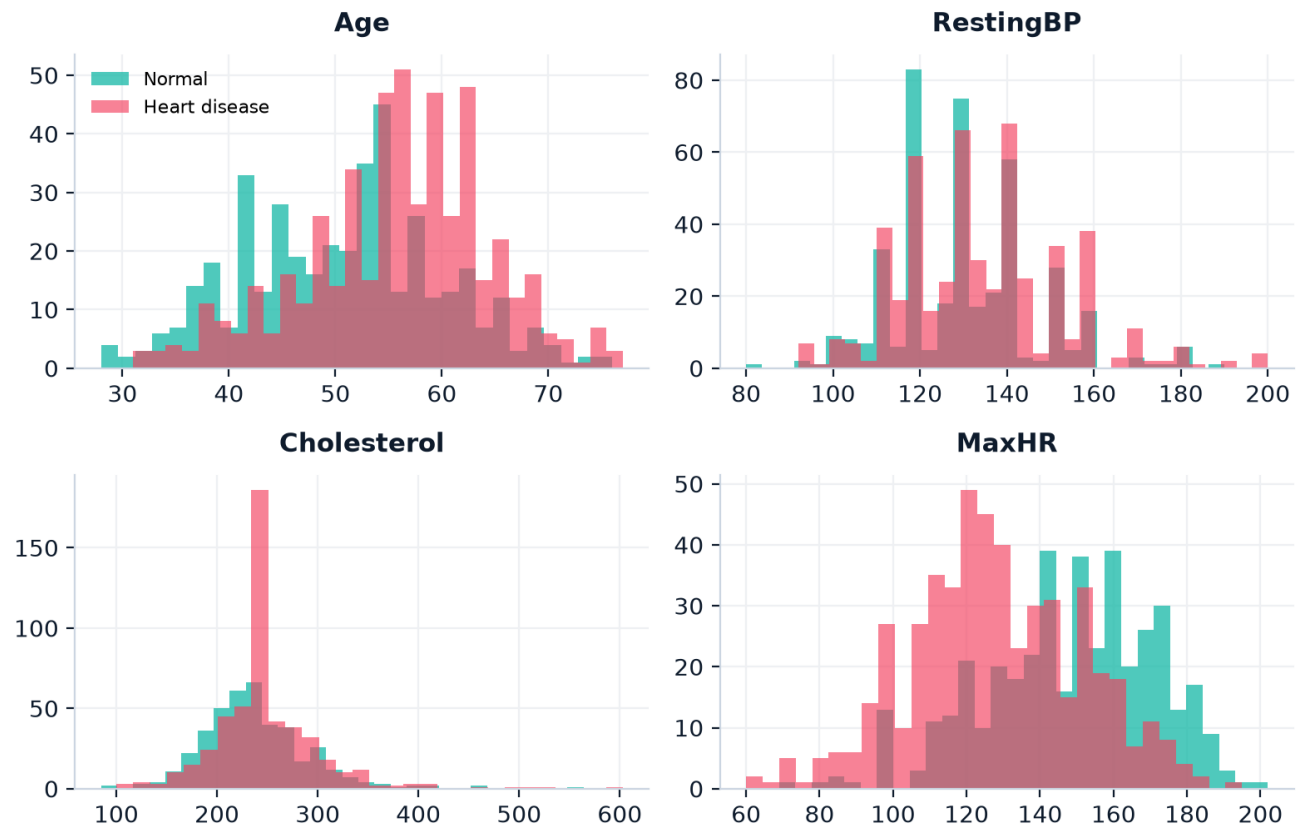


The raw histogram shows a large spike at zero (18.7% of records). After replacing zeros with the mean of non-zero values (244.64 mg/dl), the distribution becomes a plausible clinical shape. Resting BP had a single zero, replaced with 132.54 mm Hg.

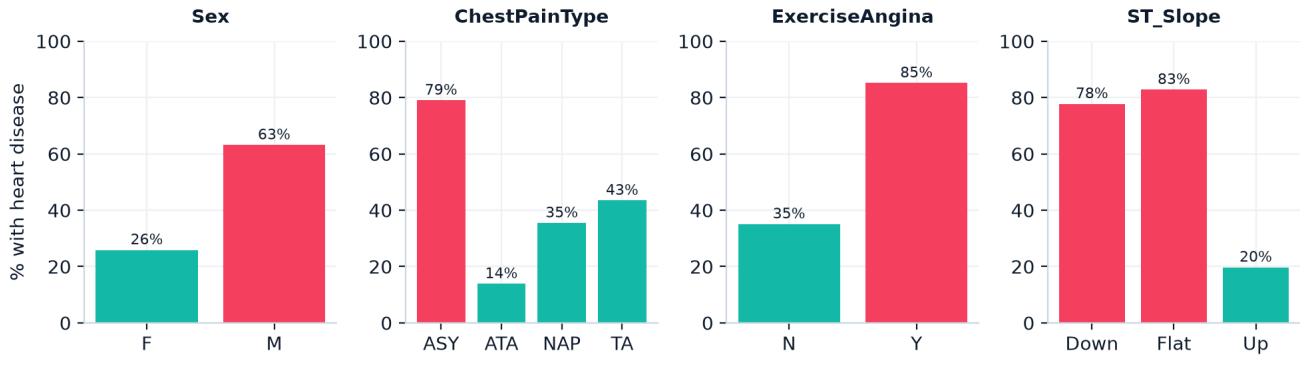
Trade-off: mean imputation creates a new spike at 244.6 and slightly hides the fact that these readings were missing. A median or model-based imputation, or a "was missing" flag, could be explored in future work.

Exploratory Analysis

Numeric features split by outcome. Overlapping teal (normal) and red (heart disease) histograms reveal where the two groups differ.

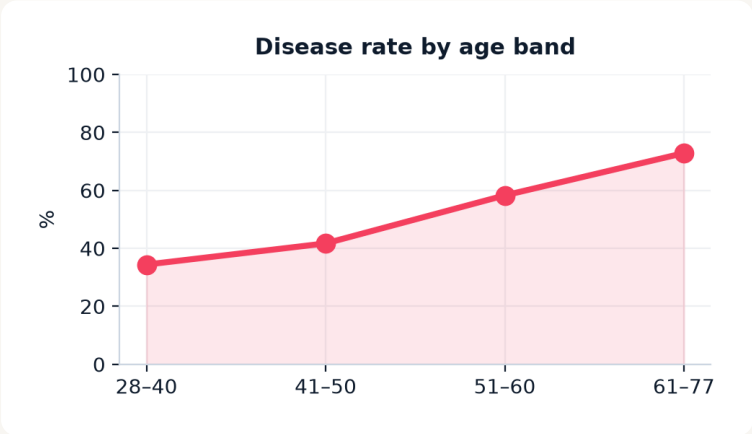


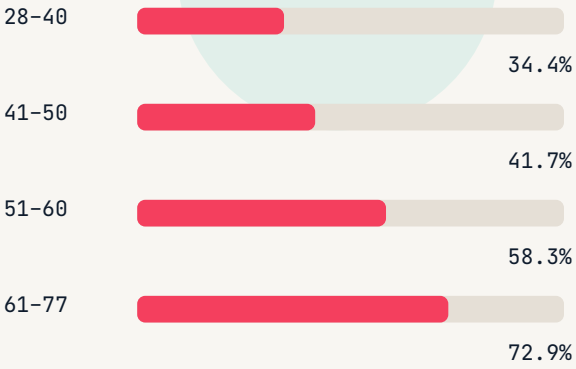
Disease rate by category



Men show a 63.2% rate vs 25.9% for women. **Asymptomatic** chest pain (79.0%) is far more dangerous than "classic" pain types — silent disease is common. Flat/Down ST slopes and exercise angina all push rates above 75%.

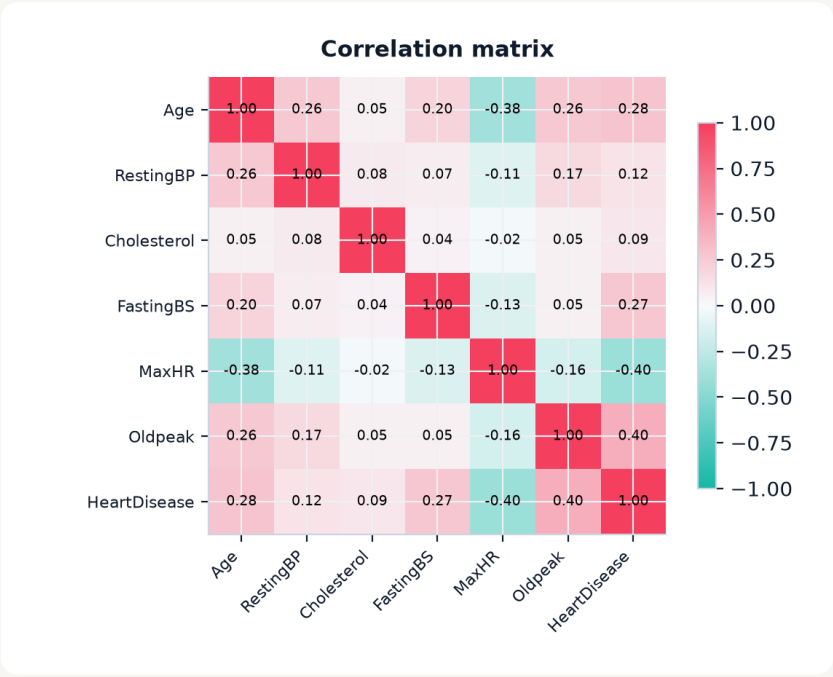
Relationships & Correlation





Risk roughly **doubles** from the youngest to the oldest band.

Correlation with the target



Oldpeak	+0.40
MaxHR	-0.40
Age	+0.28
FastingBS	+0.27
RestingBP	+0.12
Cholesterol	+0.09

Oldpeak (+0.40) and **MaxHR** (−0.40) are the strongest numeric signals. A heart that cannot reach a high rate under stress is a warning sign. Cholesterol is surprisingly weak (+0.09), partly due to the imputed zeros.

The ML Pipeline

STEP 01
Load
heart.csv · 918×12

STEP 02
Clean
Impute zero chol / BP

STEP 03
Explore
Plots + correlation

STEP 04
Encode
get_dummies, drop_first → 15 features

STEP 05
Split
80/20 · 734 / 184 · seed 42

STEP 06
Scale
StandardScaler on train only

STEP 07
Train
5 classifiers

STEP 08
Select
Best accuracy + F1

STEP 09
Export
joblib .pkl files

Encoded model inputs (15 columns)

`Age` `RestingBP` `Cholesterol` `FastingBS` `MaxHR` `Oldpeak` `Sex_M` `ChestPainType_ATA` `ChestPainType_NAP` `ChestPainType_TA` `RestingECG_Normal` `RestingECG_ST` `ExerciseAngina_Y` `ST_Slope_Flat` `ST_Slope_Up`

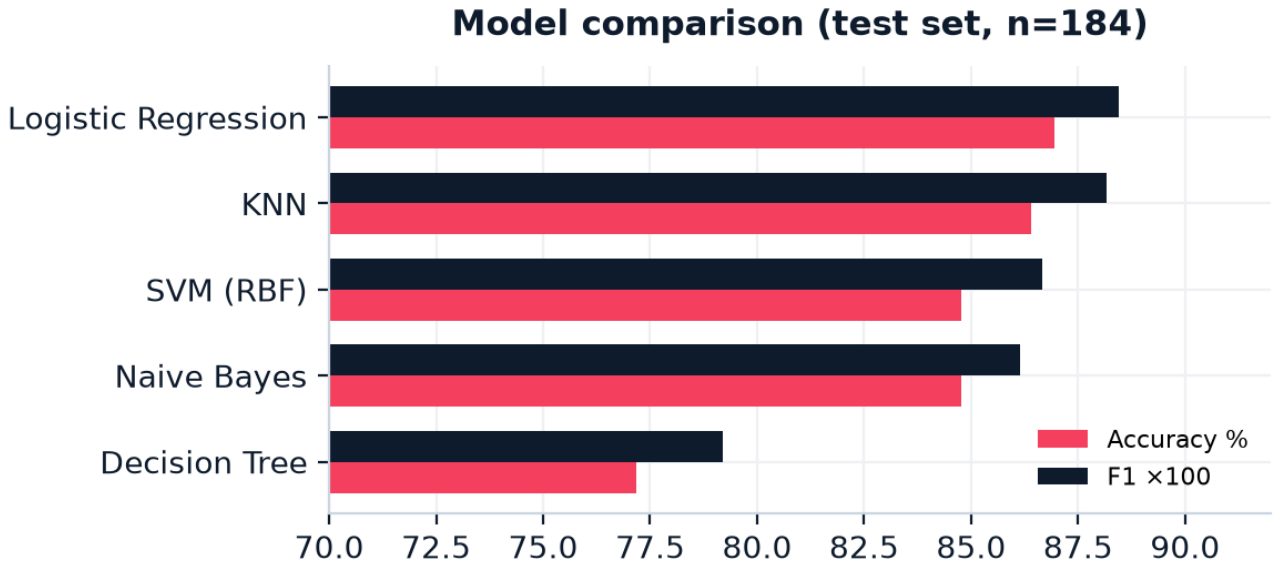
Why scale after splitting?

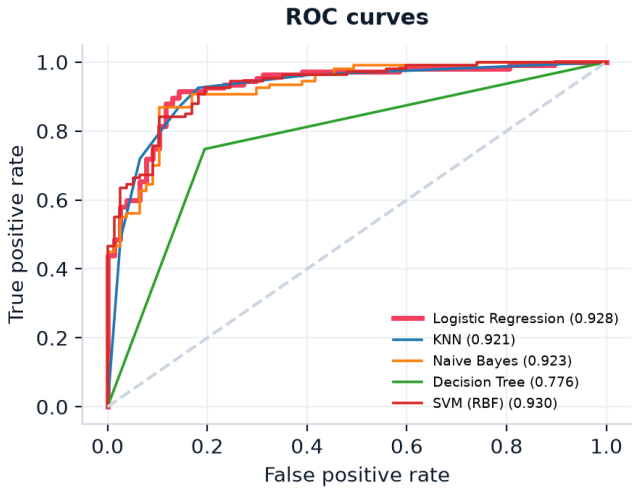
The scaler is fitted on the training set only and then applied to the test set. This prevents **data leakage** — the test patients stay truly "unseen", so the reported accuracy is an honest estimate of real-world performance.

Saved artifacts: `LR_model_heart.pkl` (the model), `scaler.pkl` (feature scaling) and `columns.pkl` (exact column order). The app loads all three so new inputs are processed identically to training data.

Model Comparison

RANK	MODEL	ACCURACY	F1	PRECISION	RECALL	ROC-AUC
1	**Logistic Regression**	**86.96%**	**0.885**	**0.911**	**0.860**	**0.928**
2	KNN	86.41%	0.882	0.894	0.869	0.921
3	Naive Bayes	84.78%	0.861	0.916	0.813	0.923
4	SVM (RBF)	84.78%	0.867	0.883	0.850	0.930
5	Decision Tree	77.17%	0.792	0.842	0.748	0.776

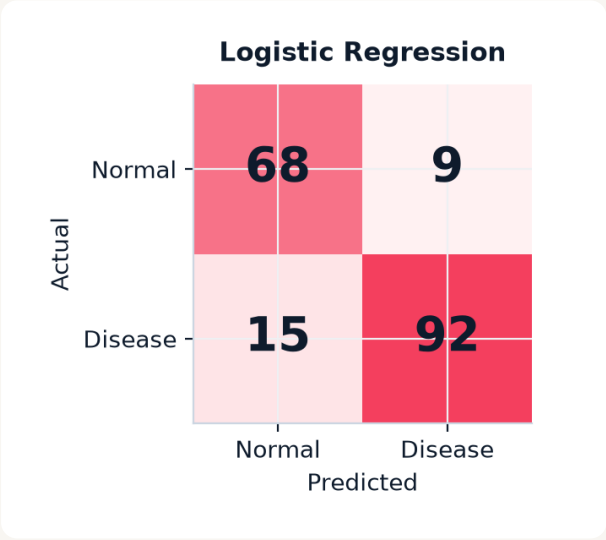




Logistic Regression has the best accuracy and F1. SVM edges it on ROC-AUC (0.930 vs 0.928), but the gap is tiny and LR is simpler, faster and explainable.

Decision Tree trails clearly — a single unpruned tree overfits. Its exact score varies run-to-run (README run: 79.89%; this rerun: 77.17%).

Inside the Winning Model



92 true positives — disease caught

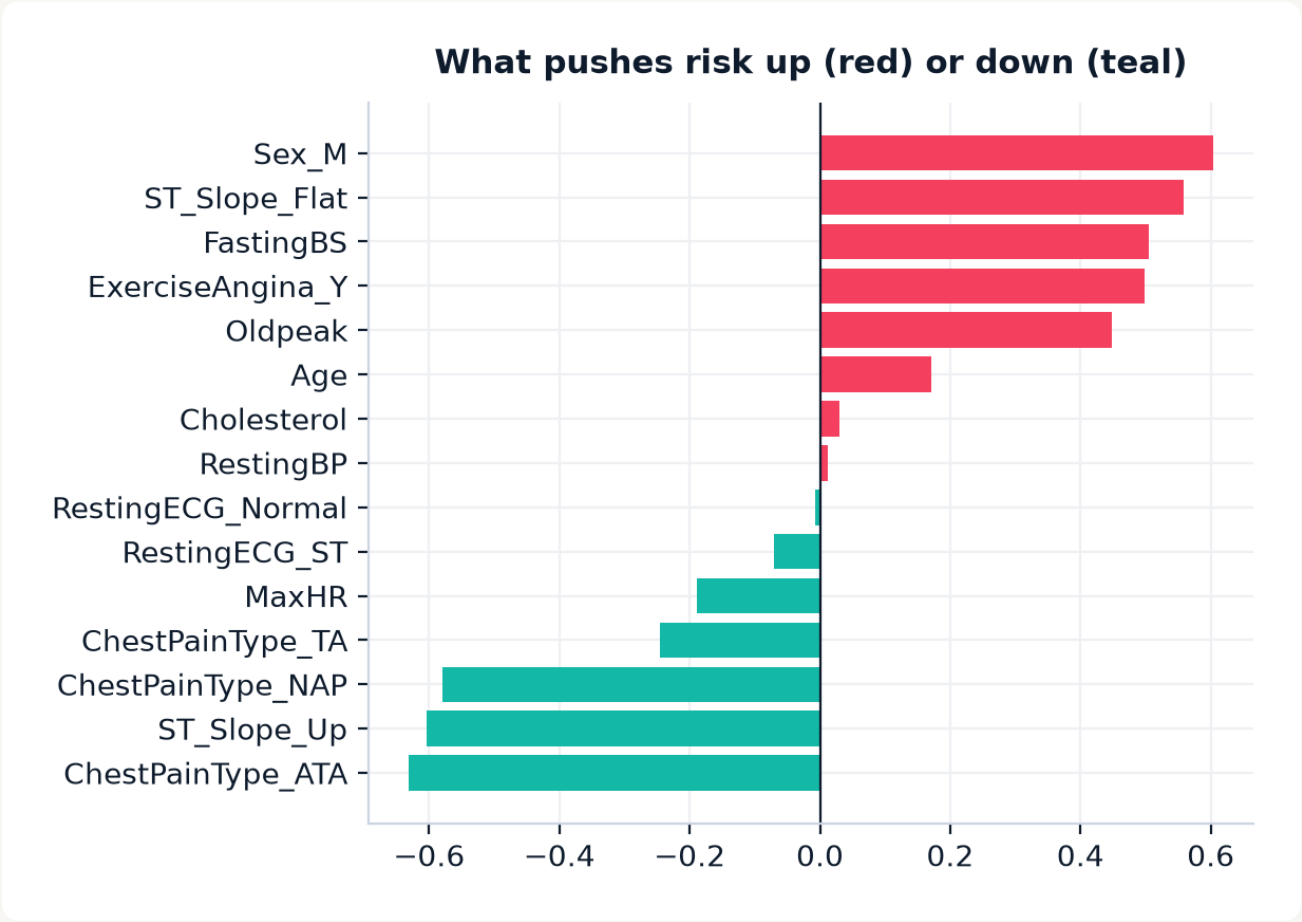
68 true negatives — correctly cleared

15 false negatives — missed cases

9 false positives — false alarms

Of 107 sick patients in the test set, the model correctly flags **92 (86.0% recall)**. When it says "high risk", it is right **91.1%** of the time.

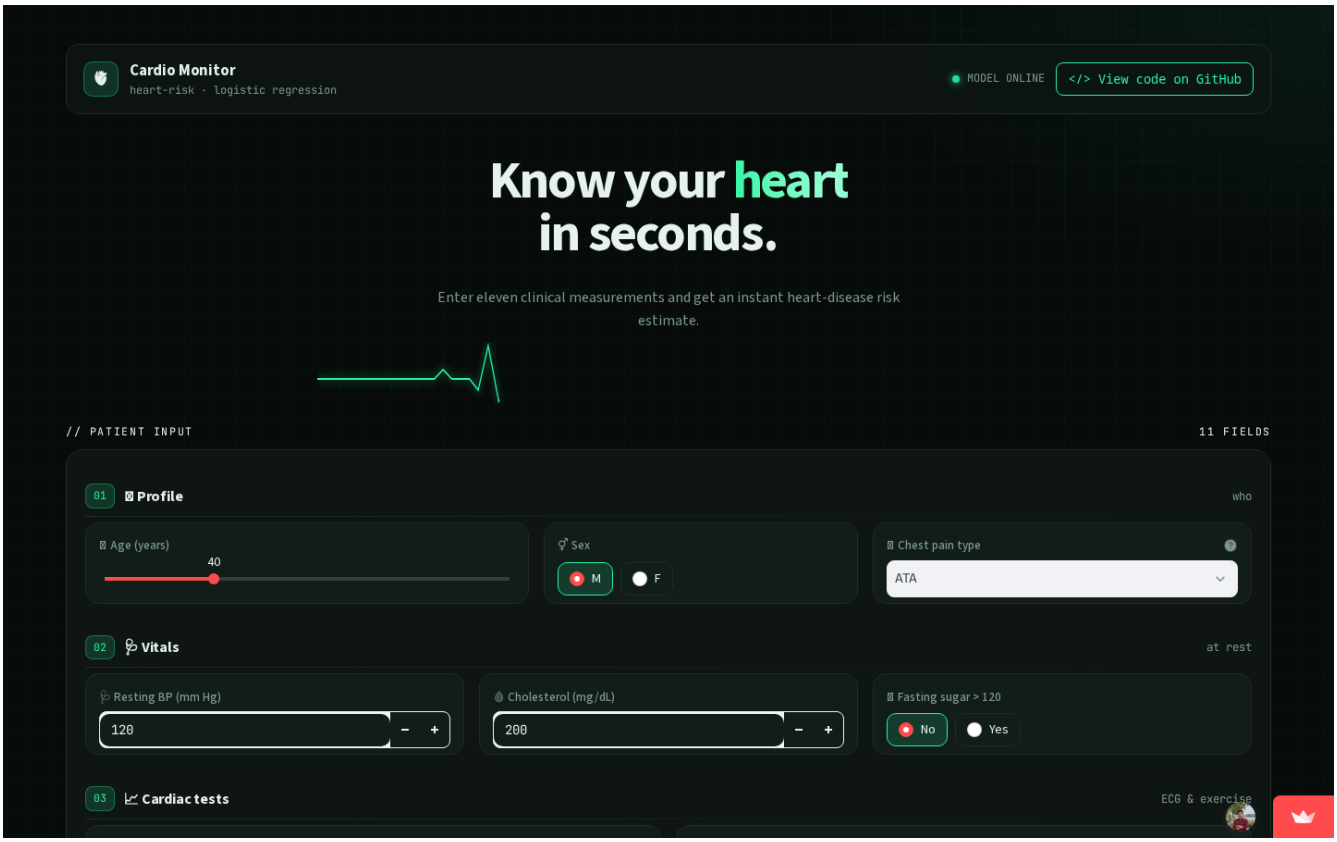
What drives the prediction



Coefficients on scaled inputs. **Flat ST slope, being male, exercise angina, fasting blood sugar and Oldpeak** raise the risk; an **Up-sloping ST segment** and non-asymptomatic chest pain types (ATA, NAP) lower it. These directions match clinical intuition — a strong sign the model learned real patterns.

The Cardio Monitor App

The trained model is served through a **Streamlit** web app styled as a dark clinical monitor with an animated ECG header. Live at **cardio-monitor-ml.streamlit.app**.



Features

- **Instant prediction**
  Risk from 11 attributes in one click
- **Guided input**
  Tabs: Profile · Vitals · Cardiac Tests
- **Visual result**
  SVG risk gauge + vital readouts
- **Fast**
  Model & scaler cached on load
- **Robust**
  Graceful message if model files missing
- **Consistent**
  Same encoding & column order as training

Tech stack

`Python` `pandas` `NumPy` `scikit-learn` `seaborn` `Matplotlib` `joblib` `Streamlit`

Conclusions & Next Steps

This project demonstrates a complete, honest ML workflow: spotting and fixing data-quality issues, exploring clinical patterns, comparing five algorithms fairly, and deploying the most reliable and explainable one as a usable tool.

What worked

- Careful handling of impossible zero values before modelling.
- Leak-free scaling and a fixed random seed for reproducibility.
- A simple, interpretable model that matches or beats complex alternatives.
- A polished, guided front-end that makes the model accessible to anyone.

Limitations

- 918 patients from combined sources — modest size and may not represent every population.
- A single 80/20 split; cross-validation would give more stable estimates.
- 15 of 107 sick test patients were missed — in medicine, false negatives are costly.
- Not a medical device: results support, never replace, a clinician.

Roadmap

01
Validate
k-fold CV + hyper-parameter tuning

02
Boost
Random Forest, XGBoost, LightGBM

03
Recall
Tune threshold to catch more cases

04
Explain
SHAP values per prediction

86.96% accurate. 92% precise. Fully explainable.

Early warning, built from routine clinical measurements.